Fleet Management & Live Bus Tracking — Production Implementation Plan v2.0

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1. Executive summary

This plan delivers a Fleet Management & Live Bus Tracking System for an initial fleet of about 100 buses in about 20 weeks (Android-first) or about 23 weeks (Android + iOS at launch). It covers scope, legal compliance, architecture, in-vehicle operations, hosting, cost, testing, rollout and handover.

Recommended approach. Confirm in Week 1 whether certified AIS-140 tracking hardware is legally required for this fleet. Then build a modular monolith, prove phone-based GPS on company-issued devices during a two-week technical spike, run a simulated 100-bus load test, pilot 3–5 buses, and roll out in batches.

What we promise and what we don't. The system will not claim uninterrupted tracking. It detects missing data, flags stale or offline buses, and shows the last known position and time.

Planning status. This is a production-ready baseline for estimation and client review. It becomes a delivery commitment once the client signs off the decision gates in Section 3, the cost sheet in Section 9, and the acceptance criteria in Section 12.

What changed from v1.0

#	Change	Section
1	AIS-140 / VLTD applicability is now a Week 1 go/no-go gate with a defined hardware path	3, 7
2	Device ownership (company-issued vs personal) is a decision gate; company-issued with device management is recommended	3, 9
3	Fake-GPS (mock location) detection and quarantine added	5

#	Change	Section
4	In-vehicle operations added: mounts, charging, data SIMs, missed trip-start alerts	9
5	Google Play location and foreground-service policy declarations added as a scheduled task	10
6	Technical spike extended from one week to two	10, 13
7	iOS timeline fixed: iOS runs in parallel, joint launch at about Week 23	13
8	Effort estimate, cost sheet, warranty and support terms added	9
9	DPDP Act 2023 and DPDP Rules 2025 named; internal wording removed	7, 17

2. Goals, scope and release boundaries

Release 1 delivers live tracking, trip management and core alerts for about 100 buses. Everything else is a later release or a separate quote.

Release 1 — MVP

- Secure login, role-based permissions, organization and fleet configuration, audit logging.
- Bus, driver, device, route, stop, schedule, assignment and trip management.
- Android driver app: trip start/end, active-trip GPS collection, visible tracking status, offline queue, mock-location detection and incident reporting.
- Live fleet map with search, status indicators, connection banner and reconnect resync.
- Location history and trip replay; stale/offline alerts; missed trip-start alerts; agreed core alerts.
- Reports with CSV export; XLSX only if confirmed for MVP.
- GPS simulator, automated tests, deployment pipeline, backups, monitoring and an operations runbook.

- **If Gate 0 finds AIS-140 hardware mandatory:** the normalized `LocationSource` interface and the first certified-device integration move from Release 2 into Release 1 (see Section 3).

Release 1 — iOS decision

If iPhone tracking is mandatory at launch, iOS is built in parallel with Android and both launch together at about Week 23. If not, Android launches at about Week 20 and iOS is quoted as a follow-on release. iOS background behavior is tested separately; Android results are never assumed to carry over.

Release 2

- Dedicated GPS hardware integration through `LocationSource` (if not already pulled into Release 1).
- Route deviation, improved ETA, document expiry alerts, push/SMS/WhatsApp notification channels.

Release 3

- Penalties, inventory/accessories, maintenance, fuel and other workflows, only after their requirements are approved.

Explicit exclusions unless separately approved

- Passenger/parent app, payroll, attendance, SMS/WhatsApp charges, paid map/routing usage above the agreed allowance, custom integrations.
- GPS hardware, phones, mounts, chargers and SIMs are **procured by the client**; this plan covers their specification, setup and testing (Section 9).

3. Decisions required before a fixed quote

The client approves these gates in writing. Gate 0 must close in Week 1; any other gate still open makes the estimate provisional.

Gate 0 — Legal tracking requirement (Week 1, go/no-go)

India's AIS-140 rules require certified Vehicle Location Tracking Devices (VLTDs) with emergency buttons in many public service vehicles. Depending on the state and permit type, this can include stage carriages, contract carriages and school buses. A phone app

does not satisfy this requirement.

The client's legal or compliance advisor confirms, in writing, which of these applies:

Outcome	What it means for this plan
A. AIS-140 not required	Phone-based tracking is the Release 1 source, as planned
B. AIS-140 required, devices already fitted	Release 1 ingests data from the existing certified devices through `LocationSource`; the driver app handles trips, incidents and backup tracking
C. AIS-140 required, devices not fitted	Client procures and fits certified devices (outside this quote); Release 1 integrates them; timeline rebaselined after vendor selection

Gates 1–10

Gate	Decision required	Why it matters
1. Device ownership	Company-issued standard phones or drivers' personal phones	Reliability, device management, spoofing risk, support cost. **Recommended: company-issued, one or two approved models, managed via Android Enterprise**
2. Fleet and users	One organization or multiple tenants; staff and concurrent users	Access boundaries, roles, load
3. Platform	Android only or Android + iOS at launch	Timeline (20 vs 23 weeks), Apple account, review lead time
4. Tracking policy	Trip-only tracking, point interval, who sees live/history data	Privacy, battery, data volume, consent
5. Retention	Raw GPS retention period and archive needs	Storage, query speed, operating cost
6. Map/routing	Tile provider, usage limits, routing source, licensing	Recurring cost, legal compliance

Gate	Decision required	Why it matters
7. Alerts	Triggers, thresholds, recipients, acknowledgement workflow	Prevents noisy or disputed alerts
8. Service levels	Support hours, uptime target, incident response, RPO/RTO	Hosting design and support staffing
9. Distribution	Managed Google Play / Play Store / private APK; TestFlight / App Store	Accounts, policy review, release process
10. Budget and ownership	Monthly hosting ceiling, cloud/app account owner, post-launch support owner	Avoids hidden costs and ownership disputes

4. Recommended technical architecture

A TypeScript modular monolith on PostgreSQL + PostGIS handles 100 buses comfortably, with clear module boundaries so parts can be split out later if measured load requires it.

Layer	Recommendation	Implementation notes
Repository	TypeScript monorepo, pnpm workspaces	Shared contracts, one CI pipeline; Turborepo optional
Backend	Node.js LTS + Fastify	Modular monolith with separate modules and service boundaries
Validation/API	Zod + OpenAPI	Validate at the boundary; generate a typed API client
Database	PostgreSQL + PostGIS	Relational constraints, spatial queries, monthly partitions for GPS history
DB access	Kysely or parameterized SQL + versioned migrations	Explicit transactions for history insert + latest-location update
Location sources	`LocationSource` interface	Normalizes phone and certified-device points into one format; required in R1 if Gate 0 = B or C

Layer	Recommendation	Implementation notes
Jobs	pg-boss (Postgres-backed)	Status sweeps, exports, retention; add Redis only on measured need
Real-time	Socket.IO	Authorized connections and organization rooms; snapshot on load and after reconnect
Web dashboard	React + Vite + TypeScript	TanStack Query/Table, React Hook Form + Zod
Map	MapLibre GL JS	Licensed tile provider with clear attribution and usage limits
Driver app	React Native + Expo development builds	Android foreground service (type: location); iOS tested separately; never validate GPS in Expo Go
Device management	Android Enterprise (MDM)	Locks battery optimization off, pins the app, pushes updates, blocks mock-location apps (if Gate 1 = company-issued)
Local queue	SQLite + secure token storage	Queued points survive network loss and app restarts
Files	S3-compatible object storage	Private, signed URLs, size/type limits, retention and backup
Delivery	Docker + CI/CD + staging	Tagged production deploys; reviewed migrations
Monitoring	Structured logs, error reporting, uptime checks, DB metrics	Alerts on ingestion failures, queue growth, stale buses, disk, backups, failed jobs

5. Data flow, data model and GPS correctness

Every point is queued on the device, validated on the server, stored once, and only moves a bus forward in time.

Data flow

1. Driver starts an assigned trip. The app checks permissions, battery optimization and mock-location status, then shows that tracking is active.
2. The app records each point with device UTC time, coordinates, accuracy, speed/heading where available, a client-generated point ID, trip/device IDs, a mock-location flag and health data (battery, charging, network).
3. Points are written to the local queue before upload. Uploads are authenticated, validated, rate-limited and size-limited.
4. The backend inserts valid points idempotently and updates the bus's latest-location row only if the point is newer.
5. The backend computes freshness, evaluates alert rules and sends authorized real-time updates to that organization.
6. The dashboard receives live events; after a reconnect it fetches a fresh snapshot, then resumes live updates.
7. After network recovery, queued points upload in order; the UI separates recovered history from current live status.

Limitation. If a phone is off, has no GPS, loses permission, or the app is force-stopped, no new points arrive. The system shows the last update time and a stale/offline state; it never implies the bus is still live.

Data model rules

- Keep `latest_bus_locations` (one row per bus) separate from append-only `trip_locations` history.
- Store timestamps as UTC `timestamptz`; display in IST. Keep both `recorded_at` and `received_at`.
- Update latest location only when the incoming `recorded_at` is newer; delayed uploads never move a bus backwards.
- Deduplicate with a unique constraint on the client-generated point ID (fallback: `device_id + recorded_at + source`).
- Validate coordinate ranges, timestamp skew, accuracy, speed/heading bounds, active trip/device assignment and payload size. Quarantine suspicious points; never silently treat them as truth.
- Partition history by month; create the next partition before month-end; alert on partition-job failure; test retention and archival.

- Enforce organization scope on every tenant query; use PostgreSQL row-level security as a second layer, not a substitute for authorization tests.
- Soft-deactivate business records; keep audit records per the approved retention policy; restrict access to driver personal data.

Fake-GPS (mock location) protection

- **Device check:** the app reads Android's mock-location flag on every point and sends it to the server.
- **Server check:** flagged points, and points implying impossible movement (for example, jumps faster than 120 km/h between points, or a fixed position while speed reports movement), are quarantined.
- **Alert:** repeated flagged points on a trip raise a "Possible location spoofing" alert to the supervisor.
- **Prevention:** on company-issued devices, device management blocks developer options and mock-location apps.
- **Audit:** quarantined points are kept for review and never shown as the bus's live position.

Tracking status

State	Starting rule	UI behavior
Fresh	Last valid point ⩽ 60 s old, trip active	Live position + last-update time
Stale	Last valid point 60 s – 5 min old	Stale warning; position not shown as live
Offline	Last valid point > 5 min old during an active trip	Offline alert + last known position and time
Not started	Scheduled trip not started within 10 min of departure time	Missed trip-start alert to supervisor
Not on trip	No active or scheduled trip	No offline alert unless the client asks for one

These thresholds are starting values, confirmed with the client and tuned during the pilot. A 15-second status sweep is the initial design; alert deduplication and resolution are tested.

6. Capacity and storage planning

At one point every 5 seconds, 100 buses produce 20 points/second: about 576,000 points per 8-hour day and about 210 million points per year.

Scenario	Points/day	Points/year
100 buses, 8 h/day, 5 s interval	576,000	~210 million
100 buses, 24 h/day, 5 s interval	1,728,000	~631 million

- Size storage from measured row and index size on realistic data, not from coordinate sizes alone.
- Run the 100-bus simulator at the agreed interval, including duplicates, delayed batches and reconnect bursts.
- Measure write and query latency, CPU, memory, disk IOPS, table/index growth, vacuum behavior and partition maintenance.
- Fix the retention policy before launch (Gate 5): keep high-resolution points for the approved period, then archive or delete. History is never unlimited.
- Bound every dashboard query by organization, bus and time range, with indexes and pagination.
- Treat 20 points/second as the baseline, not the peak. Test burst uploads after a fleet-wide outage (for example, 100 buses each uploading 10 minutes of queued points at once) and keep headroom.

7. Security, privacy and compliance

No pilot starts with an unresolved critical security finding, a failed cross-tenant test, or an undocumented tracking purpose.

Security

- TLS everywhere; secrets in a secret manager, never in source control, logs or client bundles.
- Argon2id password hashing or an established identity provider; short-lived access tokens; rotating, revocable refresh sessions; secure mobile token storage.
- Least-privilege roles and organization scoping on every endpoint, WebSocket room, export and file URL.

- Rate limits and payload/schema validation on authentication and ingestion endpoints.
- Device binding: each driver app instance is registered to a known device; unregistered devices cannot submit points.
- Append-only audit trail for logins, security events, user/role changes, trip changes, exports, configuration changes and record edits.
- Dependency and secret scanning in CI; encrypted backups; quarterly access review.

Privacy — Digital Personal Data Protection Act, 2023

Driver location is personal data under the DPDP Act, 2023 and the DPDP Rules, 2025. The client is the Data Fiduciary; the delivery team acts as a Data Processor under a written agreement.

- **Purpose and notice:** collect location only during active trips, for stated purposes (safety, operations, compliance). Drivers receive a clear notice in English and their preferred language.
- **Consent or legitimate use:** the client's advisor confirms the lawful basis (consent or the employment-related legitimate use). Consent, where used, is recorded in-app with a timestamp.
- **Minimization:** no tracking outside trips unless approved in writing; no contacts, photos or unrelated phone data.
- **Retention and erasure:** follow Gate 5; delete or anonymize when the purpose ends.
- **Breach response:** documented process to notify the Data Protection Board and affected people within the timelines set by the DPDP Rules.
- **Grievance contact:** named contact for drivers' access, correction and grievance requests.

Vehicle tracking compliance

AIS-140 / VLTD obligations are decided at Gate 0 (Section 3). The plan never assumes a phone app replaces certified hardware where the law requires it.

Exit criteria before pilot

- [ ] No unresolved critical security findings
- [ ] All cross-tenant access tests pass (HTTP, WebSocket, export, file URLs)
- [ ] No production secrets in repositories or logs
- [ ] Driver notice, lawful basis and tracking purpose documented and approved

8. Hosting and operations

The recommended launch setup is one application server plus a separate managed PostgreSQL/PostGIS database, with off-site encrypted backups and a separate staging environment.

Hosting options

Option	Setup	Trade-off
A. Recommended	App VPS (about 4 vCPU / 8 GB RAM) + managed PostgreSQL/PostGIS with automated backups + object storage	Database backups and patching handled by the provider; app server is still a single point of failure
B. Budget	Single VPS running app and database (about 4 vCPU / 8 GB / 160 GB SSD)	Lowest cost; client accepts single-server risk and in-house DB administration in writing
C. High availability	Multi-zone app servers + managed HA database	Higher cost; only if the uptime target in Gate 8 requires it

Sizes are starting points, confirmed by the 100-bus soak test. High availability is never claimed for options A or B.

Operations gates

- Daily backup status checked automatically; backups encrypted, off-site and access-restricted.
- Documented restore drill before launch and at least quarterly after launch.
- Alerts on disk, database connections, failed jobs, ingestion errors, queue age, API latency and stale/offline buses.
- Documented deploy/rollback, migration procedure, incident escalation, credential rotation and recovery contacts.
- Monthly spend alert and a written process for cost spikes.
- Separate staging with separate credentials; destructive migrations never tested on production.

9. In-vehicle operations, effort and cost

Phone-based tracking only works if every bus has a mounted, powered, managed phone; this section makes that a planned deliverable, not an assumption.

In-vehicle setup (per bus)

Item	Specification	Owner
Phone	Company-issued, one or two approved Android models, Android 12+, 4 GB+ RAM	Client procures; delivery team approves models after the spike
Mount	Vibration-rated dashboard or windshield mount with clear sky view	Client procures and fits
Power	Hardwired 12/24 V USB charger with inline fuse, fitted by the fleet electrician	Client
Data SIM	Any plan with 1 GB+/month (tracking uses well under 100 MB/month; confirmed in pilot)	Client
Device management	Enrolled in Android Enterprise: app pinned, battery optimization off, auto-updates, mock-location blocked	Delivery team configures; client owns the account
Spares	5–10% spare phones pre-enrolled for same-day swap	Client

Operational controls

- **Missed trip start:** a supervisor alert if a scheduled trip isn't started within 10 minutes of departure.
- **Device health panel:** battery, charging status, app version, permission state and last-seen time per bus.
- **Unplugged alert:** a warning when a phone on an active trip stops charging and drops below 30% battery.

- **Driver checklist:** a 30-second pre-trip check in the app (phone mounted, charging, GPS fixed) before "Start Trip" is enabled.
- **Training:** a one-page driver guide in English and the drivers' language, plus a 30-minute session per depot.

Effort estimate

Role	Android-first (20 weeks)	Android + iOS (23 weeks)
Backend developer (1.0 FTE)	20 person-weeks	23 person-weeks
Frontend developer (1.0 FTE)	17 person-weeks	19 person-weeks
Mobile developer (1.0 FTE; +0.5 FTE for iOS)	20 person-weeks	23 + 8 person-weeks
QA / DevOps (0.5 FTE)	10 person-weeks	11.5 person-weeks
Tech lead / project manager (0.25 FTE)	5 person-weeks	6 person-weeks
Total	**72 person-weeks**	**90.5 person-weeks**

Cost sheet (to complete before the fixed quote)

Cost line	Type	Basis	Amount (₹, excl. GST)
Development	One-time	Person-weeks above × agreed weekly rate	*to be filled*
Contingency	One-time	15% of development, released only through change control	*to be filled*
Hosting (Option A/B/C)	Monthly	Provider quote for chosen region and plan, incl. backups and transfer	*to be filled*
Map tiles / routing	Monthly	Provider plan at expected dashboard usage	*to be filled*
Monitoring / error reporting	Monthly	Chosen plan	*to be filled*

Cost line	Type	Basis	Amount (₹, excl. GST)
Google Play developer account	One-time	Current fee, verified at quotation	*to be filled*
Apple Developer Program	Annual	Current fee, only if iOS selected	*to be filled*
Phones, mounts, chargers, SIMs, spares	Client-procured	Per-bus spec above × 100 + spares	*client budget*
Warranty	Included	60 days of free defect fixes after go-live	Included
Support and maintenance (AMC)	Monthly / annual	Support hours, response times and updates per Gate 8	*to be filled*

No price is a fixed quote until provider, region, plan, taxes, backups, traffic, map usage and renewal prices are confirmed. GST applies as per current rates.

10. Mobile GPS technical spike and store policy

The spike runs for two weeks (Weeks 2–3) and ends in a written go/no-go on phone-based tracking before full app development starts.

Android test matrix

- At least four real devices: the approved fleet model(s), plus Xiaomi/Redmi, Oppo/Vivo/Realme and Samsung, including one low-memory device.
- 4-hour locked-screen run with the app in the background and the foreground-service notification visible.
- 10-minute network outage, queue persistence, reconnect upload, app process kill, device reboot, GPS off, permission revoked, low battery, mock-location app active.
- Record battery use, missing-point percentage, upload latency, queue growth, recovery behavior and mock-location detection rate.

iOS test matrix (if in launch scope)

- At least two real iPhones, including an older supported model.

- "Always" location permission, background mode, locked screen, network outage, force-quit, reboot, TestFlight distribution.
- Document platform limits and driver instructions; reliable tracking after a force-quit is never promised.

Pass criteria

Measure	Target	How it is counted
Missing points, 4-hour locked-screen run	< 2%	Expected points (run time ÷ interval) minus points accepted by the server, ÷ expected
Outage recovery	100% of queued points uploaded after a 10-minute outage	Points captured during outage vs accepted after reconnect
Reboot recovery	Tracking resumes without driver action, or the driver is prompted within 1 minute	Time from boot to first accepted point or prompt
Battery	≤ 8% per hour while charging is off	Battery log over the 4-hour run
Mock-location detection	100% of points flagged while a mock app is active	Flagged points ÷ points sent during mock test

Eventual delivery and real-time visibility are reported separately. If Expo-based tracking fails, evaluate a commercial background-geolocation library and confirm its licence cost; use a native Kotlin module only if that fallback is justified and tested.

Google Play and App Store policy (scheduled task)

Google Play requires declarations for apps that use location in a foreground service, and a separate review if background location permission is requested. These reviews can take days to weeks.

- **Week 3:** decide whether the app needs background location permission or only a location-type foreground service started while the app is visible (preferred: smaller review).

- **Week 8:** client-owned Play Console account ready; Data Safety form, foreground-service declaration and (if needed) background-location declaration with demo video submitted on an internal-testing build.
- **Distribution:** company-issued devices can receive the app through managed Google Play (private app), which avoids a public listing but still follows Play policy.
- **iOS (if in scope):** Apple Developer account by Week 8; location purpose strings and App Review notes prepared; TestFlight from Week 15.

11. API and real-time contracts

All endpoints are versioned, schema-validated, tenant-checked and documented in OpenAPI.

- `POST /v1/ingest/locations` accepts authenticated batches: device ID, trip ID, client point IDs, recorded timestamps, coordinates, accuracy, mock flag, and optional speed/heading/health fields.
- It returns per-point **accepted**, **duplicate** or **rejected** with stable reason codes. The app removes only accepted or duplicate points from its queue.
- Batches start at up to 200 points, with explicit request-size limits; both are tuned by testing.
- `GET /v1/live/buses` returns an authorized snapshot. Socket.IO events carry versioned payloads; every reconnect triggers a snapshot resync.
- Every endpoint has schema validation, role/tenant checks, an error contract and automated authorization tests.
- Large exports run as background jobs with filters, limits, authorization and audit logging.

Endpoint names are conventions and may change during API design; the contract stays versioned and client-approved.

12. Testing and acceptance criteria

Every test below produces recorded raw results and a sign-off; vague targets such as "zero lost points" are not used.

Area	Minimum acceptance evidence
Core workflows	Create bus/driver/route, assign and start/end a trip, see location, acknowledge an alert, export a report
GPS ingestion	20 points/s for 1 hour; p95 ingest latency < 100 ms under agreed conditions; zero duplicate history points
Soak test	100 simulated buses, 5 s interval, 8 hours; missing, duplicate and out-of-order counts plus resource metrics reported
Outage recovery	10-minute simulated outage; queued points upload after reconnect with no duplicates; UI shows stale/offline during the outage
Burst recovery	All 100 simulated buses reconnect at once with 10 minutes of queued points; backlog cleared within the agreed time with no errors
Latest location	An out-of-order point never overwrites a newer location
Spoofing	Mock-location points are flagged, quarantined and alerted; never shown as live
Missed trip start	Alert fires within 1 minute of the threshold for an unstarted scheduled trip
Tenant isolation	Automated tests prove no organization can read or modify another's data or live events
Map behavior	100 markers render and update smoothly on agreed devices; reconnect snapshot restores current state
Mobile reliability	Locked-screen, reboot, permission, battery, low-signal and app-kill tests documented on approved devices
Security	No critical unresolved findings; secret and dependency scans on; rate limits and authorization tested

Area	Minimum acceptance evidence
Recovery	Backup restored into a clean environment and data integrity verified
Pilot	3–5 real buses for 3 weeks; daily review; client signs off on update receipt rate, alert quality, battery impact and support process

13. Delivery timeline and team assumptions

Android-first launches in about 20 weeks; Android + iOS launches together in about 23 weeks. Both assume the team in Section 9, timely client decisions and Gate 0 outcome A. Outcome B or C is rebaselined after the hardware vendor is confirmed.

Weeks	Phase	Exit gate
1	Discovery and setup	Gate 0 signed; Gates 1–10 decided or dated; wireframes, repo, CI, device access
2–3	Technical spike	GPS spike report, 100-bus simulator, map/tile decision, ingestion prototype; **go/no-go signed**
4–6	Foundation	Auth/RBAC, organizations, audit, fleet CRUD, routes/stops, staging live
7–10	Tracking	Ingestion, partitions, idempotency, status engine, spoofing checks, live map, driver app, offline queue; Play declarations submitted (Week 8)
11–13	Trips and alerts	Trip lifecycle, replay, core alerts, missed-trip-start alert, incident reporting, device health panel
14–15	Reports and hardening	Exports, authorization suite, soak and burst tests, security review, restore drill, in-vehicle kit fitted on pilot buses
16–18	Pilot	3–5 buses on real routes; fixes; pilot sign-off
19–20	Android rollout (Android-first)	Batch onboarding of ~25 buses per batch, depot training, runbook, handover

If iOS is in launch scope: an extra 0.5 FTE mobile developer builds iOS in parallel from Week 7. The pilot includes at least one iPhone. Weeks 19–21 cover iOS hardening and App Store review buffer; Weeks 22–23 are the joint Android + iOS rollout.

If one person covers backend, frontend, mobile, DevOps and QA, this timeline does not apply; re-estimate after the spike and a vertical-slice prototype.

14. Roles, handover and support

- **Backend owner:** API, database, ingestion, jobs, alerts, access control.
- **Frontend owner:** dashboard, map, routes, reports, accessibility and responsive behavior.
- **Mobile owner:** GPS, permissions, queue, spoofing checks, device management setup, store declarations, release builds.
- **QA/DevOps:** CI, automated tests, load tests, deployment, backup/restore, monitoring, runbooks.
- **Client product owner:** gate decisions, device procurement and fitting, operations workflow, pilot users, acceptance and sign-off.
- **Client legal/compliance:** Gate 0 opinion, DPDP lawful basis, driver notice.
- Weekly demo and a risk/decision log; anything outside the approved MVP is a written change request.
- **Handover package:** source repository, deployment instructions, environment variable list (not secrets), API docs, schema and migrations, admin guide, driver guide (English + local language), device setup guide, test evidence, backup/restore runbook, incident contacts, known limitations, and transfer of all accounts to the client.
- **After go-live:** 60-day defect warranty, then support under the agreed AMC (Section 9).

Definition of done: reviewed, tests pass, authorization and tenant isolation checked, loading/empty/error states handled, audit events recorded where required, API docs updated, demonstrated in staging.

15. Risks and mitigations

Risk	Early signal	Mitigation
AIS-140 hardware legally required	Gate 0 returns outcome B or C	Decided in Week 1; `LocationSource` integration pulled into R1; timeline rebaselined in writing
Background tracking gaps	Spike exceeds 2% missing points	Company-issued approved models, device management, fallback library, documented force-quit limits
GPS spoofing by drivers	Flagged or impossible-movement points	Mock detection, server checks, alerts, device management blocks
Phones unplugged or missing	Unplugged/low-battery alerts, missed trip starts	Hardwired chargers, pre-trip checklist, spares, depot follow-up
Play Store policy delay	Declaration rejected or pending	Submit by Week 8; prefer foreground-service-only design; managed Google Play
Unexpected hosting cost	DB growth, map usage or backups above estimate	Usage alerts, retention, tile limits, measured sizing, monthly review
Database slowdown	Inserts or history queries degrade in soak test	Partition/index tuning, bounded queries, archive policy
False alerts	Repeated false positives per bus/day	Accuracy filters, persistence windows, pilot tuning
Cross-tenant data exposure	Authorization test fails	Central tenant guard; tests for HTTP, WebSocket, export and file paths
Privacy complaint	Driver objection or grievance	DPDP notice, trip-only tracking, grievance contact, access controls

Risk	Early signal	Mitigation
Scope creep	Release 2/3 requests in MVP sprints	Written scope baseline and change control
Single point of failure (people)	Only one person can deploy or restore	Runbooks, peer review, paired handover, restore drill
Late client decisions	Gate open past its needed-by date	Escalate in weekly review; rebaseline dates in writing

16. First 10 working days

1. Obtain the Gate 0 (AIS-140) legal opinion and confirm Gates 1–10, owners and needed-by dates.
2. Create the monorepo, strict TypeScript, linting, formatting and shared contracts.
3. Set up CI: install, lint, type-check, tests, build and container image.
4. Create local Docker Compose with PostgreSQL/PostGIS and seed data.
5. Design and review first migrations: organizations, users/roles, buses, devices, trips, latest locations, history partitions.
6. Implement the first ingestion endpoint with validation and tests for duplicates, out-of-order and mock-flagged points.
7. Build simulator v0 for 5 buses, then scale to 100.
8. Build the smallest Android tracking prototype and start the 4-hour device tests on the candidate fleet phones.
9. Show simulated moving markers on a basic MapLibre page and shortlist licensed tile providers.
10. Draft the in-vehicle kit specification (phone, mount, charger, SIM) for client procurement.

The spike report, go/no-go and revised cost follow at the end of Week 3. No dashboard polishing happens before background GPS, offline queueing, ingestion correctness and map licensing are proven.

17. Approval checklist

This plan is ready for implementation. Production launch depends on the checks below passing and being signed off.

- [ ] Gate 0 (AIS-140) legal opinion received and outcome recorded
- [ ] Scope and release boundaries approved
- [ ] Device ownership, approved models and in-vehicle kit approved
- [ ] Android/iOS scope and distribution route approved
- [ ] Tracking policy, retention and DPDP review approved
- [ ] Cost sheet completed and budget approved
- [ ] GPS spike passed, or a funded fallback approved
- [ ] Play (and App Store) declarations approved
- [ ] 100-bus load and burst tests passed
- [ ] Hosting option, monthly ceiling and cost ownership approved
- [ ] Security and tenant-isolation tests passed
- [ ] Backup restore drill succeeded; runbook and on-call owner named
- [ ] Pilot exit criteria met and final acceptance signed
- [ ] All client accounts and production assets owned by or assigned to the client

Client-facing status: Ready for implementation planning. Production launch is conditional on the Gate 0 legal decision, GPS and device validation, load testing, security review, hosting and budget approval, and pilot sign-off.